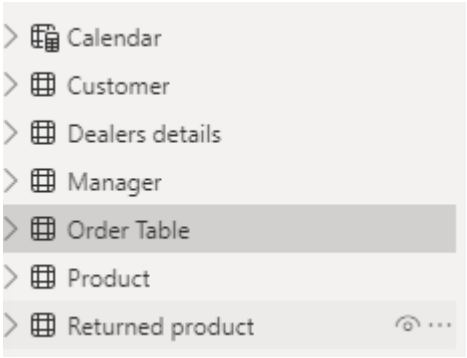
DATA SOURCE

The provider of the Data is SADDAM ANSARI from linkedin: https://github.com/user-saddam123/Car-Sales-Dashboard-New-Style-With-Graphical-Interface

DATA PROCESSING

CLEANING: Transformation of data is done firstly in excel, then loaded into power query. Further cleaning process was done to ensure a one to many relationship modelling on the order table which is the transactional table.



SPLITING INTO TABLES: Creation of different tables from the dataset. Tables created are the Demographic Table, insurance Claim analysis table and the financial analysis table. The reason for this is to have full focus on separate areas of the big data in order to get the best analysis possible out of it. The Tables are separated with the customer's ID as the primary key.

CALENDAR*:* Date, Month, Year, Day, Quarter,Month_Number

CUSTOMER: Customer id, Customer Name

DEALERS DETAILS*:* Order_id, Dealer_name, Company,Model, Transmission,Color

MANAGER*:* Manager, Region

ORDER TABLE*:* Order id, Order date, Ship date, Days interval, customer id, Destination city, Destination province, Destination Region, product id, Sales Quantity, Discount, Profit, Order_year

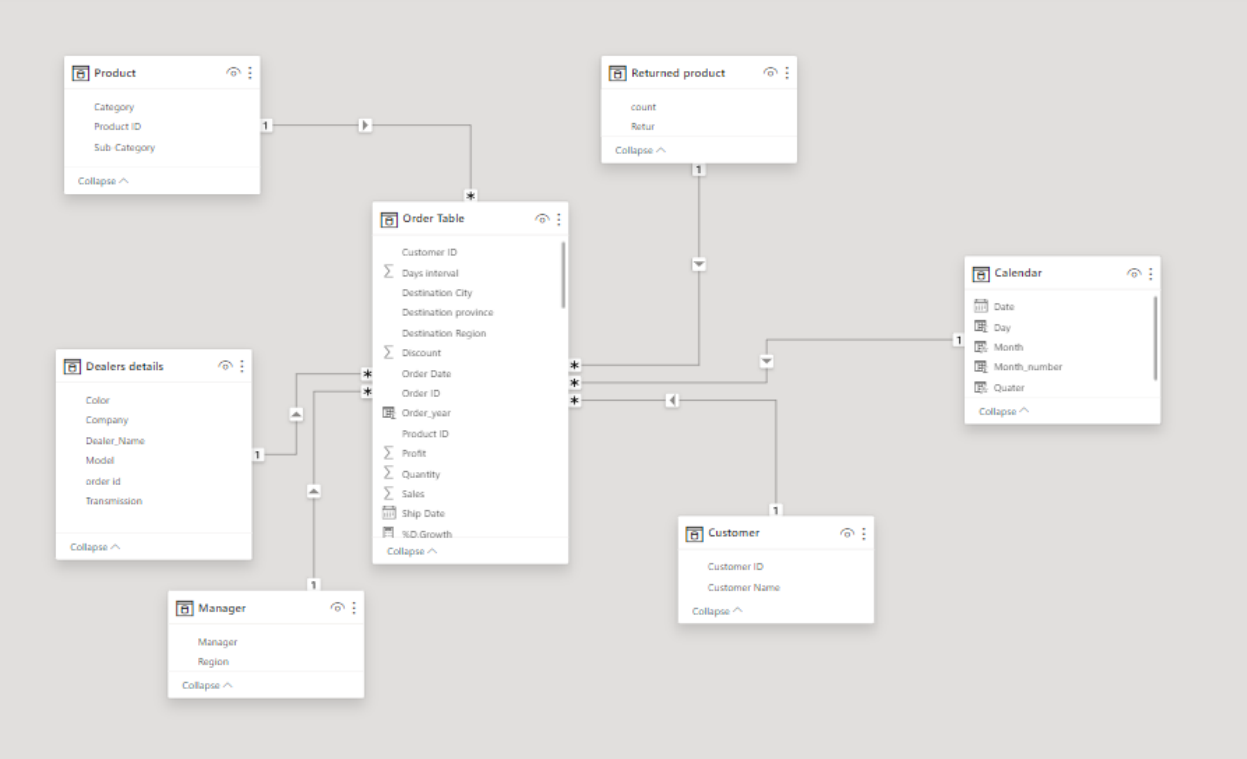
PRODUCT*:* Product id, category, sub-category

RETURNED PRODUCT*:* Id, count

MODELLING:

This modelling aspect takes a bit of engineering in to thinking. Since some of the datas are facts and needs not to be repeated over times, I carefully reduced some of the tables in to distinct with their respective IDs connected to the order

table(Transactional Table) as a one to many relationship.



CALCULATIONS AND MEASURES ON EACH TABLE

In this section, I jotted out all what's needed to be generated in terms of measure and calculated columns so as to make my work more precise, focused and timely managed. Though a few of them were also later added along during the report processing.

MEASURES AND CALCULATED COLUMNS USED AND THEIR FORMULAR.

CALCULATED COLUMNS:

#Day = 'Calendar'*[Date].[Day*]

```
#Month = 'Calendar'[Date].[Month]
#Month_number = MONTH('Calendar'[Date])
#Quater = 'Calendar'[Date].[Quarter]
#Year = 'Calendar'[Date].[Year]
#Order_year = YEAR('Order Table'[Order Date])
```

MEASURES:

```
Line measures:
#Above_Target_PC = if( [PC]>[PLY], [PC])
#Below_Target_PC = if( [PC]<[PLY], [PC])
```

```
#PC =
var _blank=ISBLANK(SELECTEDVALUE('Calendar'[Year]))
var _currentProfit= [T.Profit]
var _maxyear=MAX('Calendar'[Year])
RETURN
IF(_blank, CALCULATE([T.Profit], 'Calendar'[Year]=_maxyear),
_currentProfit
)

#PLY =
var _blank= ISBLANK(SELECTEDVALUE('Calendar'[Year]))
VAR _CurrentYear = MAX('Calendar'[Year])
VAR _PreviousYear = _CurrentYear - 1
VAR _FilteredPLY = CALCULATE([T.Profit], SAMEPERIODLASTYEAR('Calendar'[Date]))
RETURN
if( _blank, CALCULATE([T.Profit], 'Calendar'[Year]=_PreviousYear),
    _FilteredPLY
)

#Profit_highest_line = [PC]*0.25
#profit_Label = FORMAT([PC],"$#,0,.0#K")

#Subtitle_T.Profit =
VAR Growth = [%P.Growth]
VAR PLYprofit = [T.Profit(△ 'PLY- PC')]
VAR MoMLine =
IF(
    Growth > 0,
    "Growth% ▲ at " & FORMAT([%P.Growth], "+#.0%;-#.0%"),
    "Growth ▼ at " & FORMAT([%P.Growth], "+#.0%;-#.0%")
)
VAR PCLine =
IF(
    PLYprofit > 0,
    "PLY: " & FORMAT([T.Profit(△ 'PLY- PC')], "$#,0,.0#K") & " above target Profit",
    "PLY: " & FORMAT([T.Profit(△ 'PLY- PC')], "$#,0,.0#K") & " below target Profit"
)
RETURN
MoMLine & UNICHAR(10) & PCLine

#T.Profit(△ 'PLY- PC') =
var _PC = [PC]
var _PLY =[PLY]
var _Delta = _PC-_PLY
Return
_Delta

MATRIX MEASURES:
```

```
#%P.Growth&Gap = FORMAT([%P.Growth],"0%")&" | "&FORMAT([PC - PLY],"$#,0,.0#K")
#Matrix_Above_Target = IF([PC]>[PLY],[%P.Growth&Gap])

# Matrix_Above_Target_Line = IF([PC - PLY]>0,[PC - PLY])

#Matrix_Below_Target = IF([PC]<[PLY],[%P.Growth&Gap])
#Matrix_Below_Target_Line = IF([PC - PLY]<=0,[PC - PLY])
#Matrix_Up_and_Down_Color_measure =
VAR _Item = [%P.Growth]
VAR _diff = [PC - PLY]
VAR _Label =
SWITCH(
    True,
    _Item = 0 ,  "grey" ,
    _Item > 0 && _diff > 0,  "green",
    _Item < 0 && _diff < 0,  "red",
    _Item > 0 && _diff < 0,  "purple",
    _Item < 0 && _diff > 0,  "Blue"
)
RETURN
_Label
#PC - PLY = [PC]-[PLY]
GENERAL MEASURES:
#%D.Growth =

var _blank= ISBLANK(SELECTEDVALUE('Calendar'[Year]))
VAR _CurrentYear = MAX('Calendar'[Year])
VAR _PreviousYear = _CurrentYear - 1
VAR _CurrentProfit = [T.Discount]
VAR _LastYearProfit = CALCULATE([T.Discount], YEAR('Calendar'[Date]) = _PreviousYear)
VAR _Result = DIVIDE(_CurrentProfit - _LastYearProfit, _LastYearProfit)

RETURN
if( _blank, DIVIDE([DY] - [DLY], [DLY]),
    _Result
)

#%P.Growth =
VAR _CurrentYear = MAX('Calendar'[Year])
VAR _PreviousYear = _CurrentYear - 1
VAR _CurrentProfit = [PC]
VAR _LastYearProfit = CALCULATE([PC], SAMEPERIODLASTYEAR('Calendar'[Date]))
VAR _Result = DIVIDE(_CurrentProfit - _LastYearProfit, _LastYearProfit)

RETURN
    _Result
# %Q.Growth =

var _blank= ISBLANK(SELECTEDVALUE('Calendar'[Year]))
```

```
VAR _CurrentYear = MAX('Calendar'[Year])
VAR _PreviousYear = _CurrentYear - 1
VAR _CurrentProfit = [T.Quantity]
VAR _LastYearProfit = CALCULATE([T.Quantity], YEAR('Calendar'[Date]) = _PreviousYear)
VAR _Result = DIVIDE(_CurrentProfit - _LastYearProfit, _LastYearProfit)

RETURN
if( _blank, DIVIDE([QY] - [QLY], [QLY]),
    _Result
)

#%S.Growth =

var _blank= ISBLANK(SELECTEDVALUE('Calendar'[Year]))
VAR _CurrentYear = MAX('Calendar'[Year])
VAR _PreviousYear = _CurrentYear - 1
VAR _CurrentProfit = [T.Sales]
VAR _LastYearProfit = CALCULATE([T.Sales], YEAR('Calendar'[Date]) = _PreviousYear)
VAR _Result = DIVIDE(_CurrentProfit - _LastYearProfit, _LastYearProfit)

RETURN
if( _blank, DIVIDE([SY] - [SLY], [SLY]),
    _Result
)

#Customer_Trans_Count =
var _blank=ISBLANK(SELECTEDVALUE('Calendar'[Year]))
var _currentcount= COUNT('Order Table'[Customer ID])
RETURN
IF(_blank, CALCULATE(COUNT('Order Table'[Customer ID]), 'Calendar'[Year]=
MAX('Calendar'[Year])),
_currentcount
)

#Discount_Label = FORMAT([dy],"$#,0,.0#K")

#Discount_Up_and_Down_Color_measure =
VAR _Item = [%D.Growth]
VAR _Label =
SWITCH(
    True,
    _Item = 0,  "grey" ,
    _Item > 0,  "green",
    _Item < 0,  "red"
)
RETURN
_Label
```

```
#DLY =
var _blank= ISBLANK(SELECTEDVALUE('Calendar'[Year]))
VAR _CurrentYear = MAX('Calendar'[Year])
VAR _PreviousYear = _CurrentYear - 1
VAR _FilteredPLY = CALCULATE([T.Discount], SAMEPERIODLASTYEAR('Calendar'[Date]))
RETURN
if( _blank, CALCULATE([T.Discount], 'Calendar'[Year]=_PreviousYear),
    _FilteredPLY
)

#DY =
var _blank=ISBLANK(SELECTEDVALUE('Calendar'[Year]))
var _currentProfit= [T.Discount]
RETURN
IF(_blank, CALCULATE([T.Discount], 'Calendar'[Year]= 2022),
_currentProfit
)

#Gap_Up_and_Down_Color_measure =
VAR _Item = [PC - PLY]
VAR _Label =
SWITCH(
    True,
    _Item = 0,  "grey" ,
    _Item > 0,  "green",
    _Item < 0,  "red"
)
RETURN
_Label

#Last Updated Timestamp = NOW()
#Profit_Up_and_Down_Color_measure =
VAR _Item = [%P.Growth]
VAR _Label =
SWITCH(
    True,
    _Item = 0,  "grey" ,
    _Item > 0,  "green",
    _Item < 0,  "red"
)
RETURN
_Label
")
#Profit_Up_and_Down_Format_measure =
VAR _Item = [%P.Growth]
VAR _Label =
SWITCH(
    True,
```

```
    _Item = 0,  " " & " ",
    _Item > 0,  " "&"+" & FORMAT(_Item,"0.00%"),
    _Item < 0,  " "&"-" & FORMAT(_Item,"0.00%")
)
RETURN
_Label

#QLY =
var _blank= ISBLANK(SELECTEDVALUE('Calendar'[Year]))
VAR _CurrentYear = MAX('Calendar'[Year])
VAR _PreviousYear = _CurrentYear - 1
VAR _FilteredPLY = CALCULATE([T.Quantity], SAMEPERIODLASTYEAR('Calendar'[Date]))
RETURN
if( _blank, CALCULATE([T.Quantity], 'Calendar'[Year]=_PreviousYear),
    _FilteredPLY
)

#Quantity_Label = FORMAT([QY],"$#,0,.0#K")
#Quantity_Up_and_Down_Color_measure =
VAR _Item = [%Q.Growth]
VAR _Label =
SWITCH(
    True,
    _Item = 0,  "grey" ,
    _Item > 0,  "green",
    _Item < 0,  "red"
)
RETURN
_Label
#QY =
var _blank=ISBLANK(SELECTEDVALUE('Calendar'[Year]))
var _currentQuantity= [T.Quantity]
RETURN
IF(_blank, CALCULATE([T.Quantity], 'Calendar'[Year]= 2022),
_currentQuantity
)

#Sales_Label = FORMAT([SY],"$#,0,.0#K")
#Sales_Up_and_Down_Color_measure =
VAR _Item = [%S.Growth]
VAR _Label =
SWITCH(
    True,
    _Item = 0,  "grey" ,
    _Item > 0,  "green",
    _Item < 0,  "red"
)
RETURN
```

```
_Label

#scatter_chart_Up_and_Down_Color_measure =
VAR _Item = [PC]
VAR _Label =
SWITCH(
    True,
    _Item = 0,  "grey" ,
    _Item > 0,  "green",
    _Item < 0,  "red"
)
RETURN
_Label

#SLY =
var _blank= ISBLANK(SELECTEDVALUE('Calendar'[Year]))
VAR _CurrentYear = MAX('Calendar'[Year])
VAR _PreviousYear = _CurrentYear - 1
VAR _FilteredPLY = CALCULATE([T.Sales], SAMEPERIODLASTYEAR('Calendar'[Date]))
RETURN
if( _blank, CALCULATE([T.Sales], 'Calendar'[Year]=_PreviousYear),
    _FilteredPLY
)

#SY =
var _blank=ISBLANK(SELECTEDVALUE('Calendar'[Year]))
var _currentProfit= [T.Sales]
RETURN
IF(_blank, CALCULATE([T.Sales], 'Calendar'[Year]= 2022),
_currentProfit
)
#T.Discount = SUM('Order Table'[Discount])
#T.Profit = SUM('Order Table'[Profit])
#T.Quantity = SUM('Order Table'[Quantity])
#T.Sales = SUM('Order Table'[Sales])

STACK BAR MEASURES:
#Debug_selected_month = SELECTEDVALUE('Calendar'[Month])
#PLaceholder_Region = maxx(ALL('Order Table'[Destination Region]),[T.Profit])/5
#Zer0 = 0
```

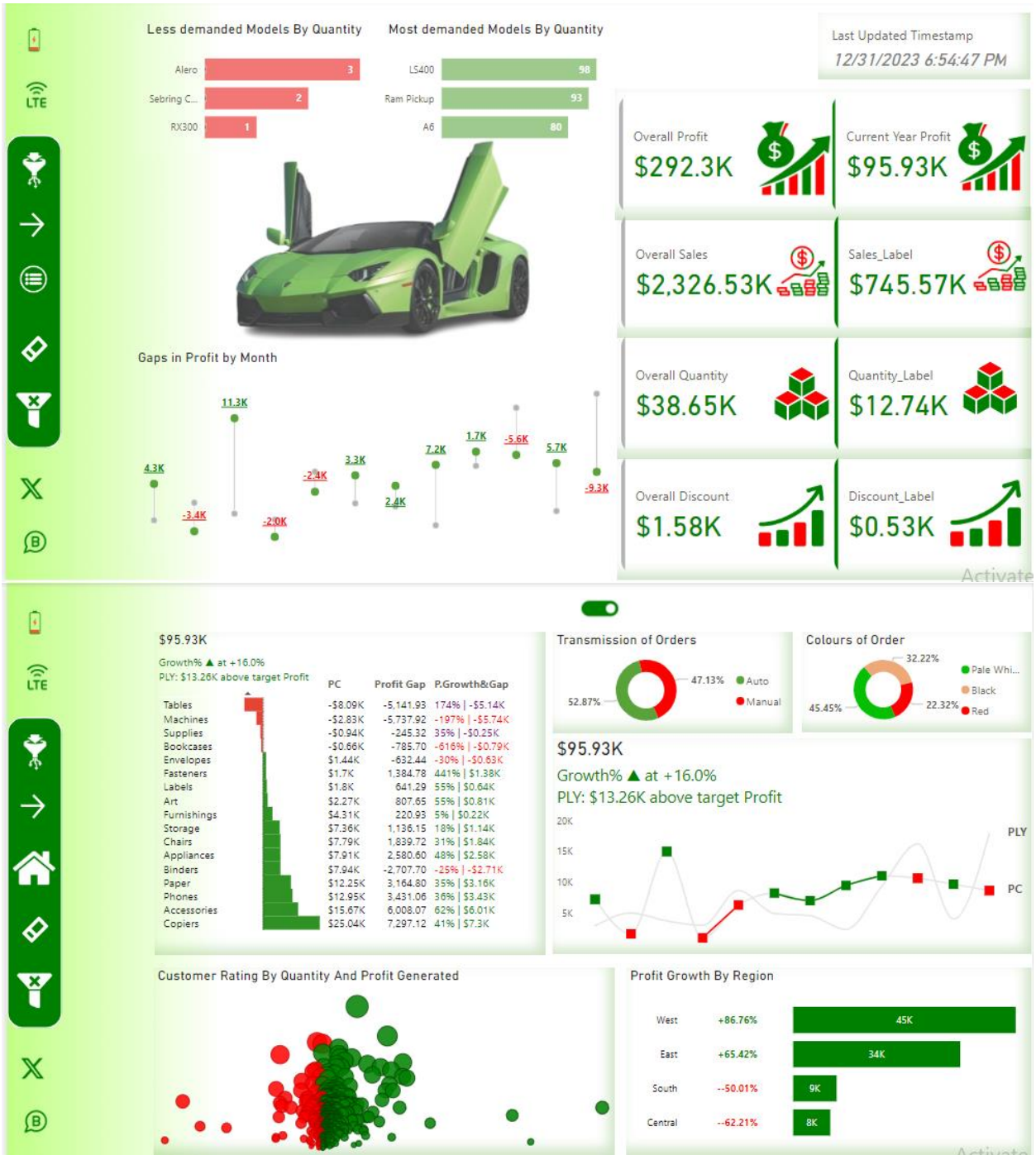
DESIGN

Colours: Green, Yellow, Red, Blue
Fuctions: DAX, Page navigation, Bookmark, tooltips,links
Others: Graphics design, icons, Buttons,

REPORT VIEW

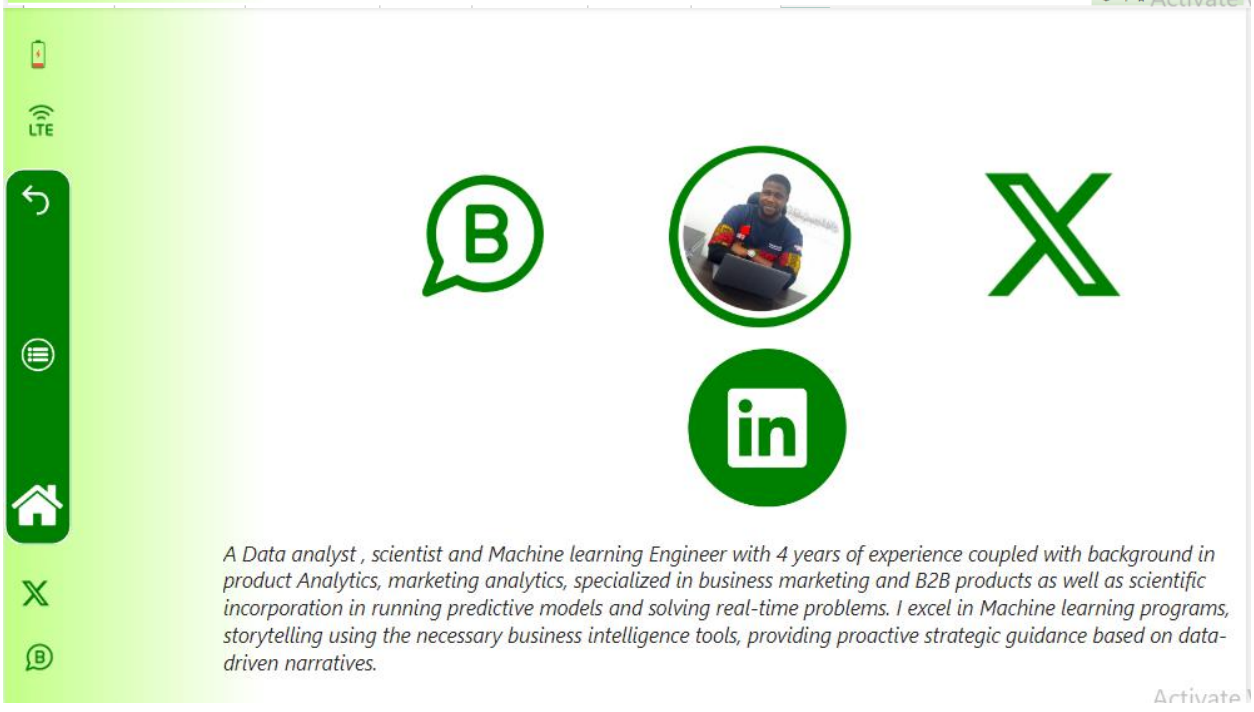
Find below the report charts…

DATA *DATASET FILE*, *VISUALIZATION FILE*, *VISUALIZATION VIEW*

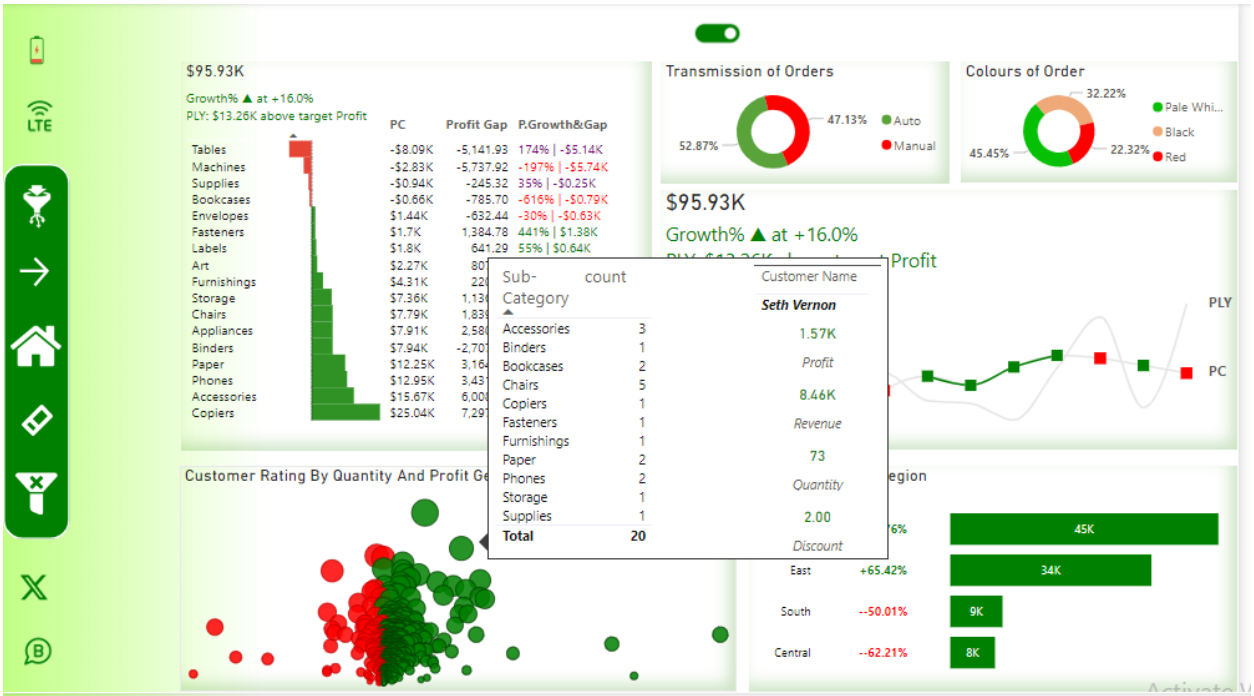


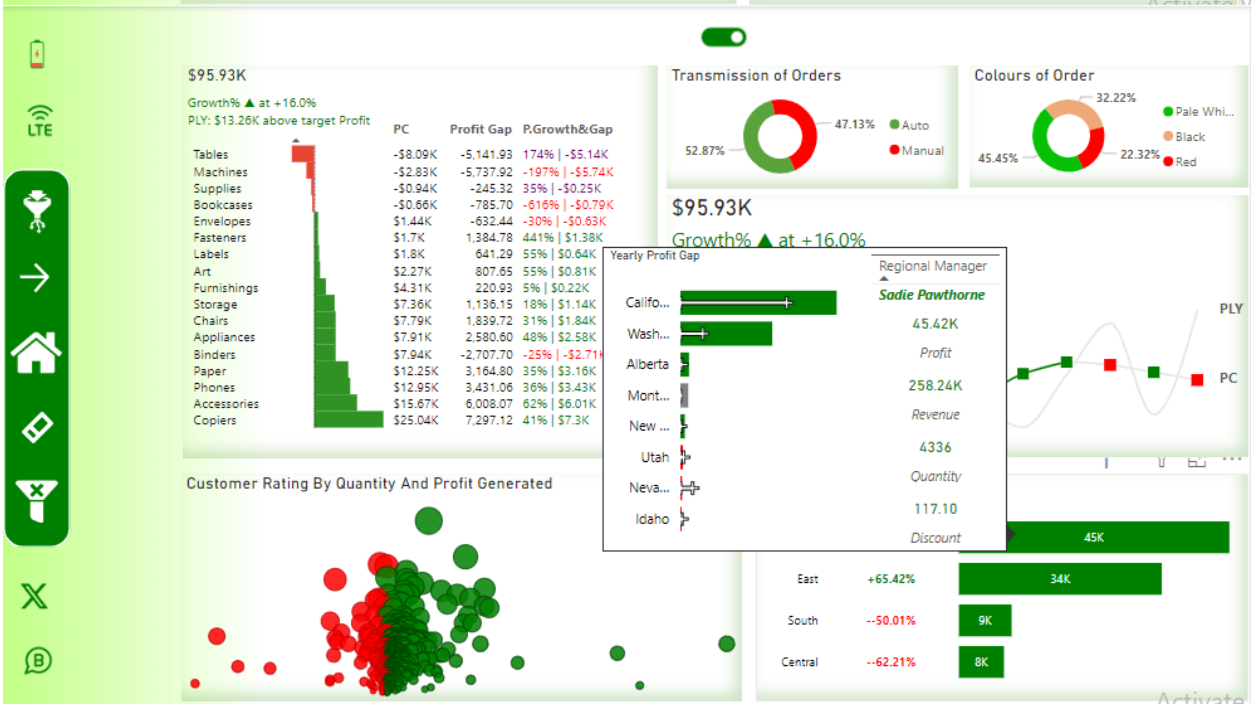


	Profit	Sales	Quantity	Discount	T.Count	P.Growth&Gap
Accessories	$15.67K	59,946.23	1079	19.80	275	62% | $6.01K
Appliances	$7.91K	43,111.16	659	29.30	168	48% | $2.58K
Art	$2.27K	9,022.70	1126	22.60	289	55% | $0.81K
Binders	$7.94K	73,651.27	2096	191.50	508	-25% | -$2.71K
Bookcases	-$0.66K	30,354.07	280	17.22	78	-616% | -$0.79K
Chairs	$7.79K	98,032.38	699	32.70	196	31% | $1.84K
Copiers	$25.04K	62,916.67	79	3.40	23	41% | $7.3K
Envelopes	$1.44K	3,378.57	241	6.20	71	-30% | -$0.63K
Fasteners	$1.7K	6,161.71	277	4.40	68	441% | $1.38K
Furnishings	$4.31K	29,927.28	1172	43.40	333	5% | $0.22K
Labels	$1.8K	3,987.03	453	7.20	116	55% | $0.64K
Machines	-$2.83K	43,887.88	122	10.10	34	-197% | -$5.74K
Paper	$12.25K	28,173.53	1708	36.60	463	35% | $3.16K
Phones	$12.95K	105,668.39	1117	48.40	299	36% | $3.43K
Storage	$7.36K	70,273.81	1037	22.50	292	18% | $1.14K
Supplies	-$0.94K	16,075.66	195	4.20	60	35% | -$0.25K
Tables	-$8.09K	60,999.20	397	27.60	106	174% | -$5.14K
Total	**$95.93K**	**745,567.53**	**12737**	**527.12**	**3379**	**16% | $13.26K**



A Data analyst , scientist and Machine learning Engineer with 4 years of experience coupled with background in product Analytics, marketing analytics, specialized in business marketing and B2B products as well as scientific incorporation in running predictive models and solving real-time problems. I excel in Machine learning programs, storytelling using the necessary business intelligence tools, providing proactive strategic guidance based on data-driven narratives.


$95.93K
Growth% ▲ at +16.0%
PLY: $13.26K above target Profit
PC
Profit Gap
P.Growth&Gap
Tables -$8.09K -5,141.93 174% | -$5.14K
Machines -$2.83K -5,737.92 -197% | -$5.74K
Supplies -$0.94K -245.32 35% | -$0.25K
Bookcases -$0.66K -785.70 -616% | -$0.79K
Envelopes $1.44K -632.44 -30% | -$0.63K
Fasteners $1.7K 1,384.78 441% | $1.38K
Labels $1.8K 641.29 55% | $0.64K
Art $2.27K
Furnishings $4.31K
Storage $7.36K
Chairs $7.79K
Appliances $7.91K
Binders $7.94K
Paper $12.25K
Phones $12.95K
Accessories $15.67K
Copiers $25.04K
Transmission of Orders
47.13% Auto
52.87% Manual
Colours of Order
32.22% Pale Whi...
Black
45.45%
22.32% Red
Sub-Category
count
Accessories 3
Binders 1
Bookcases 2
Chairs 5
Copiers 1
Fasteners 1
Furnishings 1
Paper 2
Phones 2
Storage 1
Supplies 1
Total 20
Customer Name
Seth Vernon
1.57K
Profit
8.46K
Revenue
73
Quantity
2.00
Discount
PLY
PC
Customer Rating By Quantity And Profit Ge
East +65.42%
South --50.01%
Central --62.21%
45K
34K
9K
8K


$95.93K
Growth% ▲ at +16.0%
PLY: $13.26K above target Profit
PC
Profit Gap
P.Growth&Gap
Tables -$8.09K -5,141.93 174% | -$5.14K
Machines -$2.83K -5,737.92 -197% | -$5.74K
Supplies -$0.94K -245.32 35% | -$0.25K
Bookcases -$0.66K -785.70 -616% | -$0.79K
Envelopes $1.44K -632.44 -30% | -$0.63K
Fasteners $1.7K 1,384.78 441% | $1.38K
Labels $1.8K 641.29 55% | $0.64K
Art $2.27K 807.65 55% | $0.81K
Furnishings $4.31K 220.93 5% | $0.22K
Storage $7.36K 1,136.15 18% | $1.14K
Chairs $7.79K 1,839.72 31% | $1.84K
Appliances $7.91K 2,580.60 48% | $2.58K
Binders $7.94K -2,707.70 -25% | -$2.71K
Paper $12.25K 3,164.80 35% | $3.16K
Phones $12.95K 3,431.06 36% | $3.43K
Accessories $15.67K 6,008.07 62% | $6.01K
Copiers $25.04K 7,297.12 41% | $7.3K
Transmission of Orders
47.13% Auto
52.87% Manual
Colours of Order
32.22% Pale Whi...
Black
45.45%
22.32% Red
Yearly Profit Gap
Califo...
Wash...
Alberta
Mont...
New ...
Utah
Neva...
Idaho
Regional Manager
Sadie Pawthorne
45.42K
Profit
258.24K
Revenue
4336
Quantity
117.10
Discount
PLY
PC
Customer Rating By Quantity And Profit Generated
East +65.42%
South --50.01%
Central --62.21%
45K
34K
9K
8K



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About the Analyst:

Olaniyi Micheal Oluwatosin

A Data analyst , scientist and Machine learning Engineer with 4 years of experience coupled with background in product Analytics, marketing analytics, specialized in business marketing and B2B products as well as scientific incorporation in running predictive models and solving real-time problems. I excel in Machine learning programs, storytelling using the necessary business intelligence tools, providing proactive strategic guidance based on data-driven narratives.